CUSTOMER CHURN ANALYSIS

Understanding customer retention through Python, SQL, and Power BI

PROJECT TYPE	TOOLS
Customer Analytics	Python • SQL Server • Power BI
DATASET	**PRIMARY OBJECTIVE**
Telco Customer Churn	Identify churn patterns and recommend retention actions

PROJECT REPORT

Prepared as part of a data analytics internship project

1. Executive Summary

This project analyzes customer churn in a telecommunications dataset to understand how contract type, payment method, tenure, and monthly charges relate to customer attrition. Python was used for data preparation and exploratory analysis, SQL Server for KPI queries, and Power BI for an interactive dashboard.

Metric	Result	Interpretation
Total customers	7,032	Customers represented in the reported KPI
Churned customers	1,869	Customers marked as churned
Overall churn rate	26.58%	Approximately one in four customers churned

Key message: churn is not evenly distributed. Month-to-month contracts and electronic-check payments showed notably higher churn in the reported analysis, making them useful starting points for targeted retention efforts.

2. Project Objectives

• Measure overall churn and summarize customer KPIs.
• Compare churn across contract types and payment methods.
• Explore how tenure and monthly charges relate to churn.
• Build an interactive Power BI dashboard.
• Turn findings into practical customer-retention recommendations.

3. Tools and Workflow

Python (Pandas and Matplotlib): inspected the dataset, converted TotalCharges to numeric, handled invalid/missing values, created exploratory charts, and exported the cleaned CSV.
SQL Server: created a database and queried overall KPIs, churn by contract, churn by payment method, and a targeted customer segment.
Power BI: built KPI cards, churn comparison charts, grouped churn-rate visuals, and slicers for Contract and Payment Method.

4. Data Preparation

The TotalCharges field was converted to numeric using coercion for values that could not be parsed. Rows with missing TotalCharges after conversion were removed before exporting the cleaned dataset as *cleaned_churn_data.csv*. The cleaned file was then used for SQL analysis and Power BI reporting.

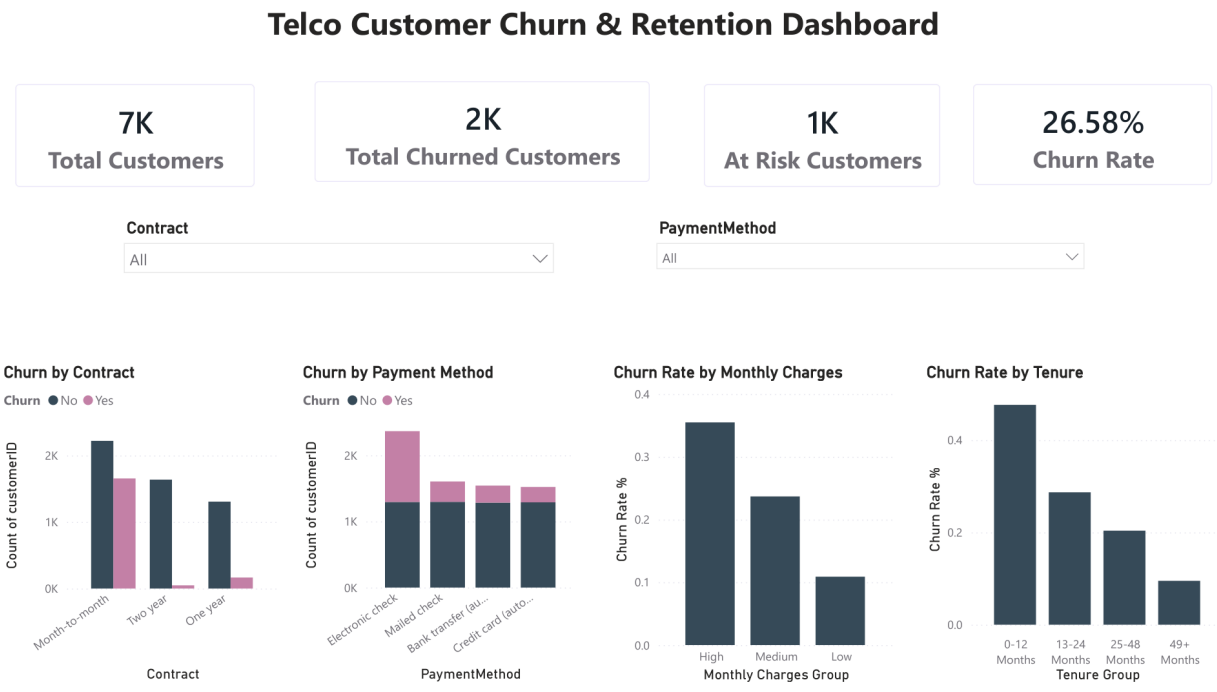
Data quality note: confirm the final row count against the exported CSV before reporting a cleaned row count separately from the KPI totals.

5. Key Findings

- **Overall churn:** 1,869 of 7,032 customers churned, corresponding to 26.58%.
- **Contract type:** month-to-month customers had a reported churn rate of 42.71%, compared with 11.28% for one-year contracts and 2.85% for two-year contracts.
- **Payment method:** electronic-check customers had a reported churn rate of 45.29%. Bank-transfer and credit-card methods showed lower churn, around 15–16% in the analysis.
- **Tenure:** churn was higher among customers in the first 12 months, suggesting that early customer experience matters for retention.
- **Monthly charges:** higher monthly-charge groups were associated with higher churn in the dashboard. This association does not, by itself, prove that price causes churn.

6. Power BI Dashboard

The dashboard contains four KPI cards (Total Customers, Total Churned Customers, At Risk Customers, and Churn Rate), churn comparisons by contract and payment method, churn rate by monthly-charge group, and churn rate by tenure group. Contract and PaymentMethod slicers are available for interactive filtering.



Dashboard screenshot provided by the project author.

7. Business Recommendations

1. **Encourage longer-term contracts.** Test a modest incentive, such as a 5–10% discount or service upgrade, for month-to-month customers who switch to one-year or two-year plans. Measure conversion and retention to assess cost-effectiveness.
2. **Promote automatic payments.** Encourage electronic-check customers to adopt credit-card or bank-transfer automatic payments. A small, time-limited credit or loyalty reward could be tested against a control group.
3. **Target relevant service gaps.** The SQL query identified 1,138 non-churned customers on month-to-month contracts without TechSupport. Consider a targeted TechSupport or OnlineSecurity trial for this segment and track subsequent churn.
4. **Improve early-tenure onboarding.** Introduce proactive check-ins during the first six months, clear setup guidance, and early troubleshooting.
5. **Monitor value perception.** Review high-charge customer groups for service usage, plan fit, and perceived value rather than assuming price alone causes churn.

8. Important Interpretation Notes

- The At Risk Customers card follows the Python definition: customers who have not churned and whose tenure is 12 months or less. This is different from the 1,138 customers selected by SQL (non-churned, month-to-month, and without TechSupport).
- Power BI rounds the cards to 7K and 2K, while the exact KPI values used in this report are 7,032 and 1,869.
- These results describe patterns in this dataset. Validate interventions with future cohorts or controlled experiments before assuming they will reduce churn.

9. Conclusion

The analysis found an overall churn rate of 26.58%, with higher churn concentrated among month-to-month customers, electronic-check users, and customers with shorter tenure. Python preparation, SQL KPI queries, and an interactive Power BI dashboard provide a repeatable approach to monitoring churn and identifying groups for retention initiatives. The next step is to test targeted interventions and measure their impact on churn, retention, and revenue.

Appendix: Deliverables Checklist

■ Python notebook or script
■ SQL script with database setup and KPI queries
■ Power BI file: *Telco_Customer_Churn_Dashboard.pbix*
■ PDF report with dashboard screenshot
■ Short project video (at least 30 seconds)
■ GitHub repository and LinkedIn project post